Four architectures

One encoder module is reused across passes in the two-pass variants.

TranAD+

2 passes · feedback

One-decoder

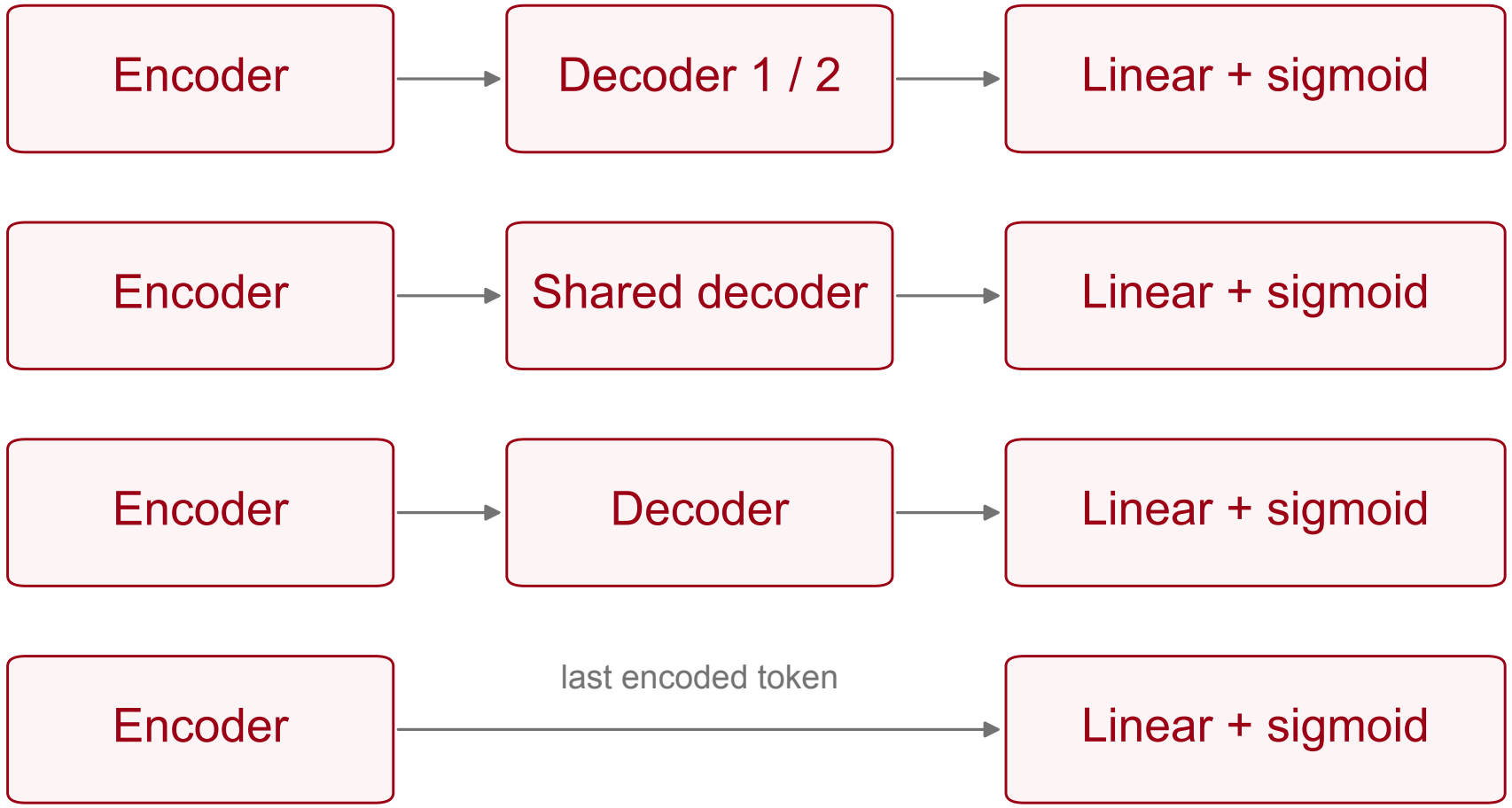
2 passes · feedback

One-pass

1 pass

Encoder-only

1 pass



Experimental grid & evaluation

3 datasets · 4 architectures · 4 W/FF settings · 5 seeds · 50 epochs

Dataset	Features	Reference	Window only	FF only	Both
OPS-SAT	1	10 / 8	5 / 8	10 / 4	5 / 4
SMAP	25	10 / 50	5 / 50	10 / 25	5 / 25
MSL	55	50 / 110	25 / 110	50 / 55	25 / 55

BF200

200 thresholds · 21 alpha values

Select the largest test F1

POT

SPOT threshold for each alpha

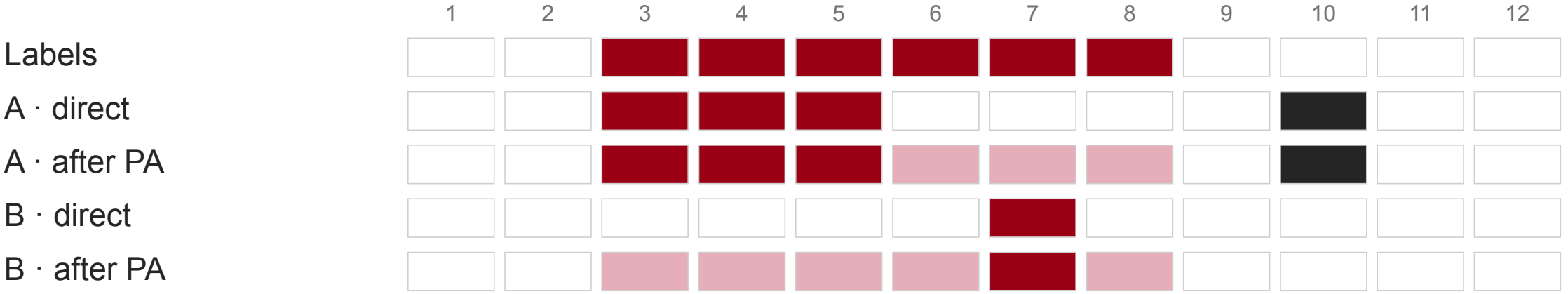
Select the largest test F1 across alpha

W / FF in table · single-output models have no alpha sweep

AUC-PA: binary predictions after PA · (Recall + Specificity) / 2

What point adjustment changes

One hit inside a labelled anomaly interval fills that interval during evaluation.



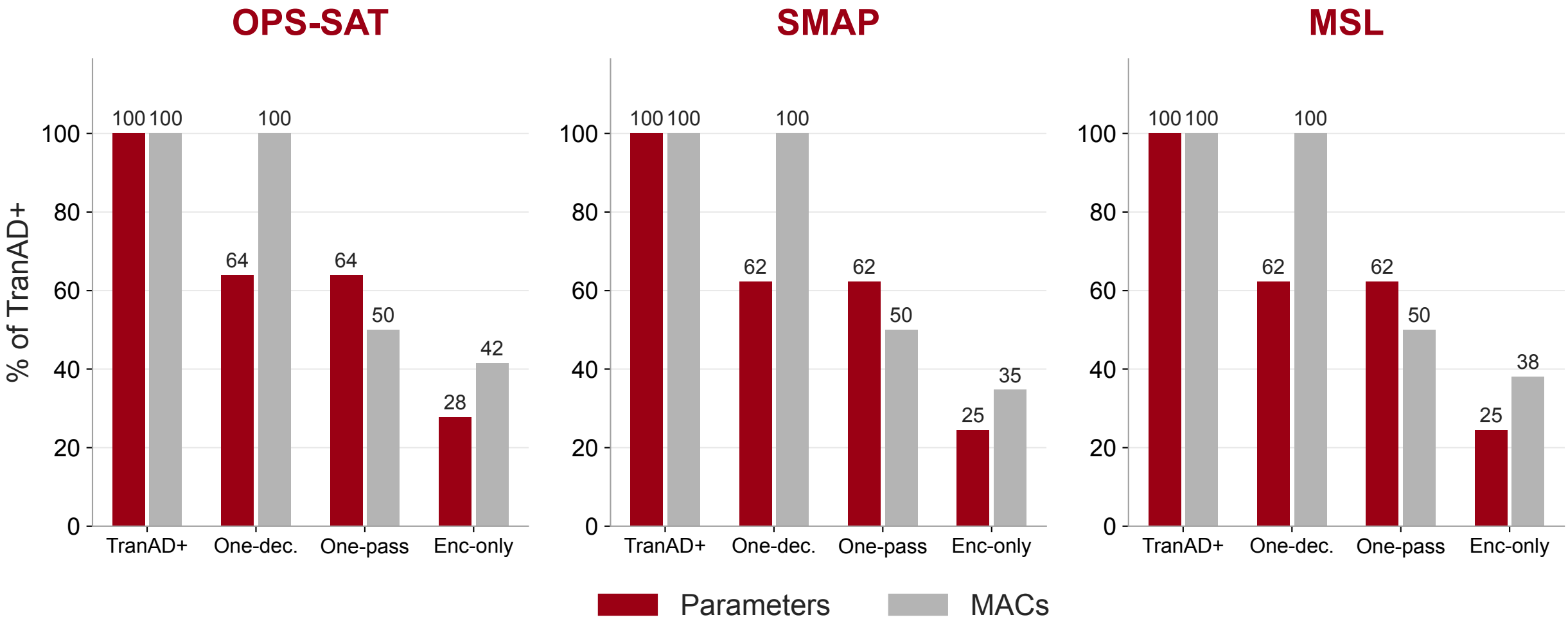
Red: detected · pale red: filled by PA · dark: false alarm

Model	Scoring	TP	FP	FN	Precision	Recall	F1
A	Direct	3	1	3	75.0	50.0	60.0
A	PA	6	1	0	85.7	100.0	92.3
B	Direct	1	0	5	100.0	16.7	28.6
B	PA	6	0	0	100.0	100.0	100.0

Intervals come from labels, not window size. False alarms outside remain unchanged.

Architecture cost

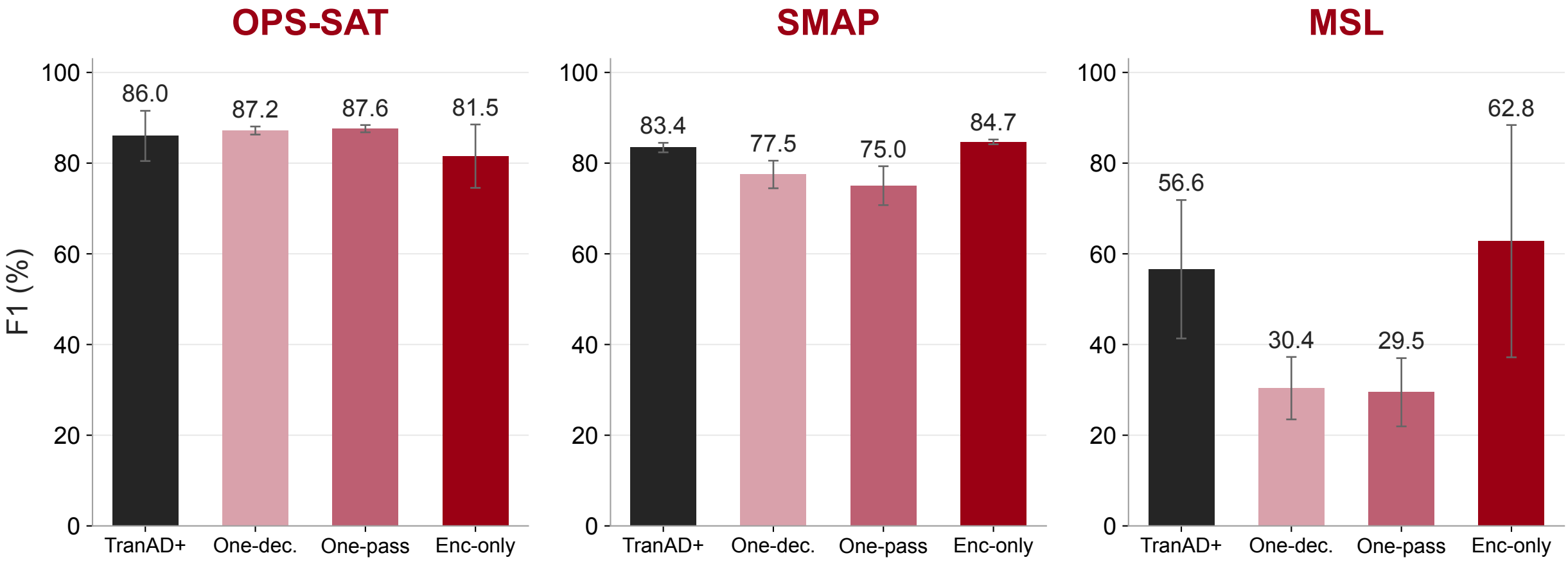
Reference W/FF · quantities relative to TranAD+ FP32



Sharing decoder weights saves parameters; removing a pass also saves computation.

Architectures · BF200

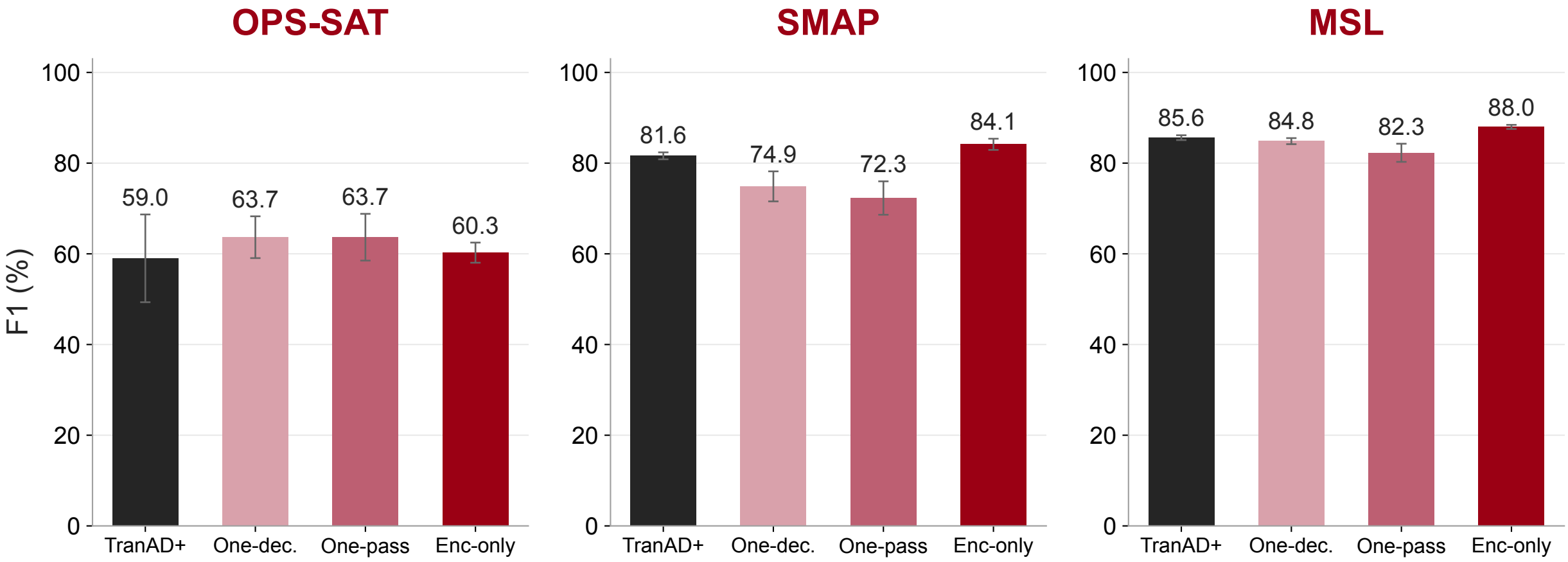
Reference W/FF · FP32 · PA · mean ± sample SD over 5 model seeds



Architectural changes have different effects across the three datasets.

Architectures · POT

Reference W/FF · FP32 · PA · mean ± sample SD over 5 model seeds



Architectural changes have different effects across the three datasets.

Precision, recall & AUC-PA

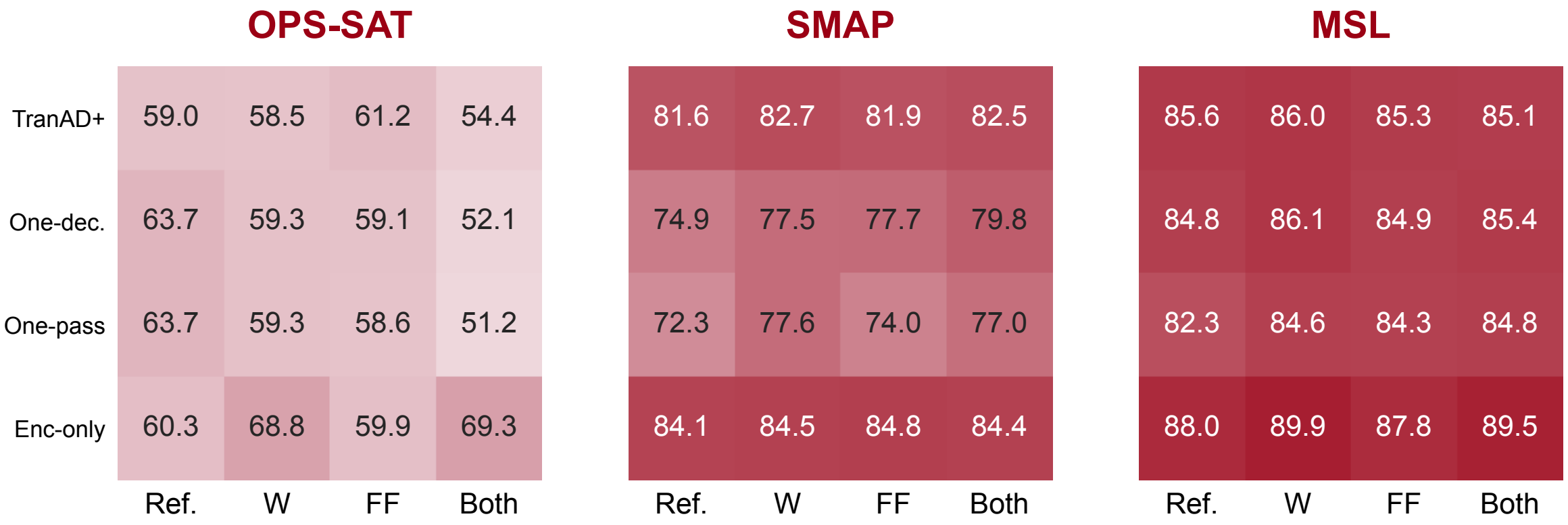
Reference W/FF · FP32 · POT + PA · mean over 5 model seeds

Dataset	Model	Precision	Recall	F1	AUC-PA
OPS-SAT	TranAD+	92.20	43.92	59.01	70.91
OPS-SAT	Encoder-only	92.38	44.74	60.26	71.30
SMAP	TranAD+	88.18	75.96	81.61	87.23
SMAP	Encoder-only	88.62	80.12	84.15	89.31
MSL	TranAD+	78.85	93.65	85.62	95.37
MSL	Encoder-only	83.90	92.51	87.99	95.23

Percentages at the selected operating point; metrics are not maximized separately.

Window & FF controls · POT

POT + PA · mean over 5 seeds in every architecture/configuration cell



Reference, reduced window, reduced FF, or both: no universal monotonic pattern.

Does more training help?

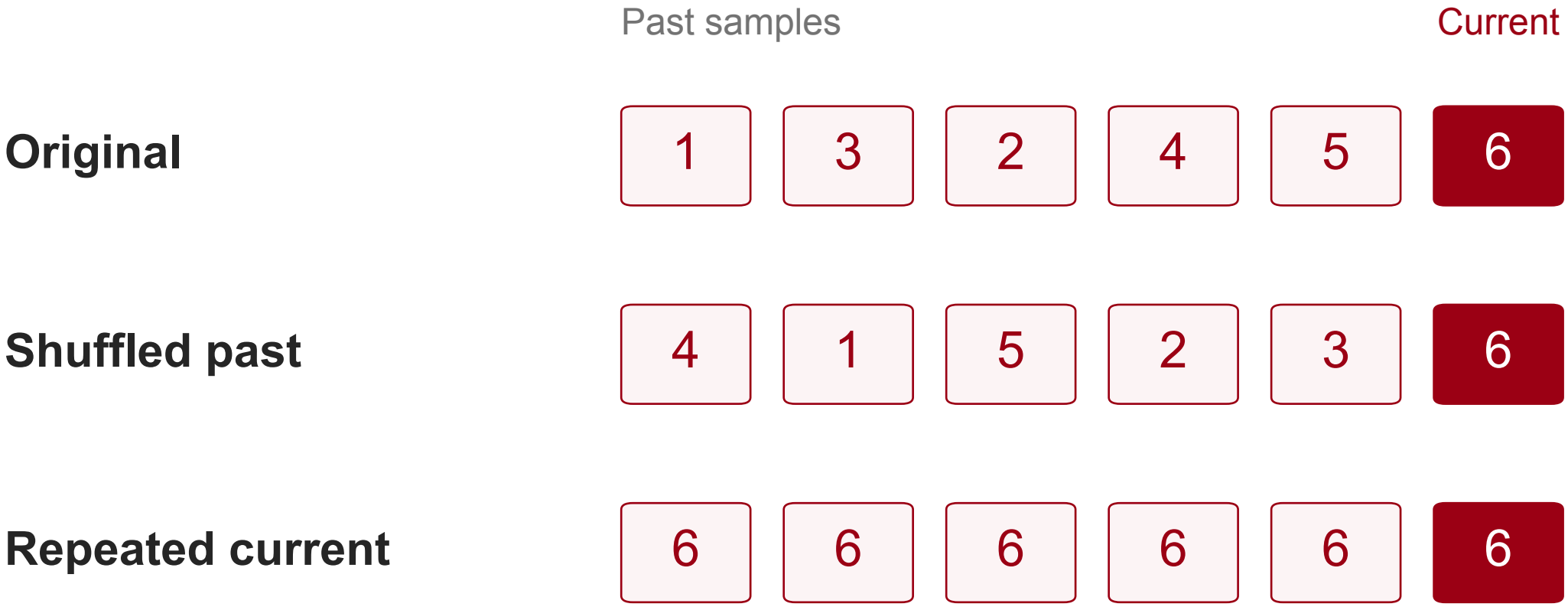
Reference W/FF · PA · mean F1 (%) over 5 seeds · continuation from epoch 50

Dataset	Model	BF 50 ep.	BF 100 ep.	POT 50 ep.	POT 100 ep.
OPS-SAT	TranAD+	86.00	85.74	59.01	60.55
OPS-SAT	Encoder-only	81.53	84.10	60.26	63.04
SMAP	TranAD+	83.43	82.64	81.61	80.68
SMAP	Encoder-only	84.68	86.15	84.15	84.96
MSL	TranAD+	56.60	45.19	85.62	84.15
MSL	Encoder-only	62.79	48.33	87.99	88.79

An additional 50 epochs do not give a uniform improvement.

Changing the past, keeping the target

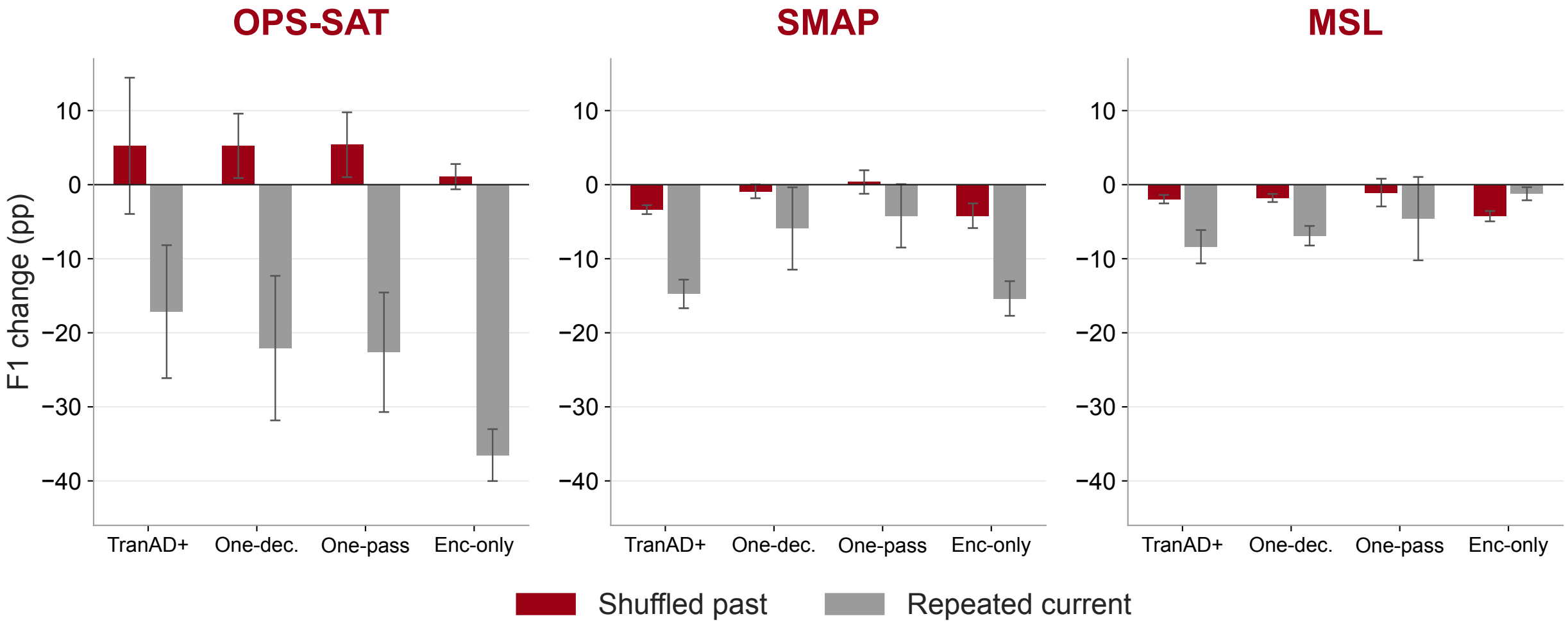
Only past samples change. The current sample and reconstruction target stay fixed.



Frozen inference tests sensitivity; matched retraining checks adaptation to the input condition.

Frozen models · context sensitivity

POT + PA · fixed parent alpha/threshold · paired changes over 5 model seeds



Shuffle results average 3 permutations per model.

OPS-SAT · matched retraining controls

OPS-SAT · POT F1 (%) · matched training/test inputs

Model	Original PA	Shuffle PA	Repeat PA	W = 1 PA	W = 1 no PA
TranAD+	59.01	57.56	61.92	36.70	9.92
One-decoder	63.66	57.43	65.23	43.31	14.48
One-pass	63.66	57.38	65.07	36.70	9.92
Encoder-only	60.26	60.26	78.01	91.32	12.64

No-PA column: same PA-selected operating point.

FF = 8 · matched training/test inputs · W = 10 except the W = 1 control.

Two quantization experiments

INT8 and INT4 · two ways to use reduced precision

Weights only

Weights saved as INT8 or INT4.

Calculations run in FP32.

Weights + calculations

Weights and inputs are integers.

Matrix products run with integers.

Softmax, sigmoid and some other steps remain in FP32.

Weight-only PTQ · F1 with PA

PA · mean over 80 parents per dataset: 4 architectures × 4 W/FF settings × 5 seeds

Dataset	Method	FP32 (%)	INT8 (%)	INT4 (%)
OPS-SAT	BF200	85.55	85.21	83.61
OPS-SAT	POT	59.88	60.03	66.18
SMAP	BF200	81.38	81.37	80.34
SMAP	POT	79.82	79.61	76.73
MSL	BF200	46.71	46.71	46.69
MSL	POT	85.91	85.98	86.24

Weight-only sensitivity; floating-point matrix products.

Integer inference · F1 with PA

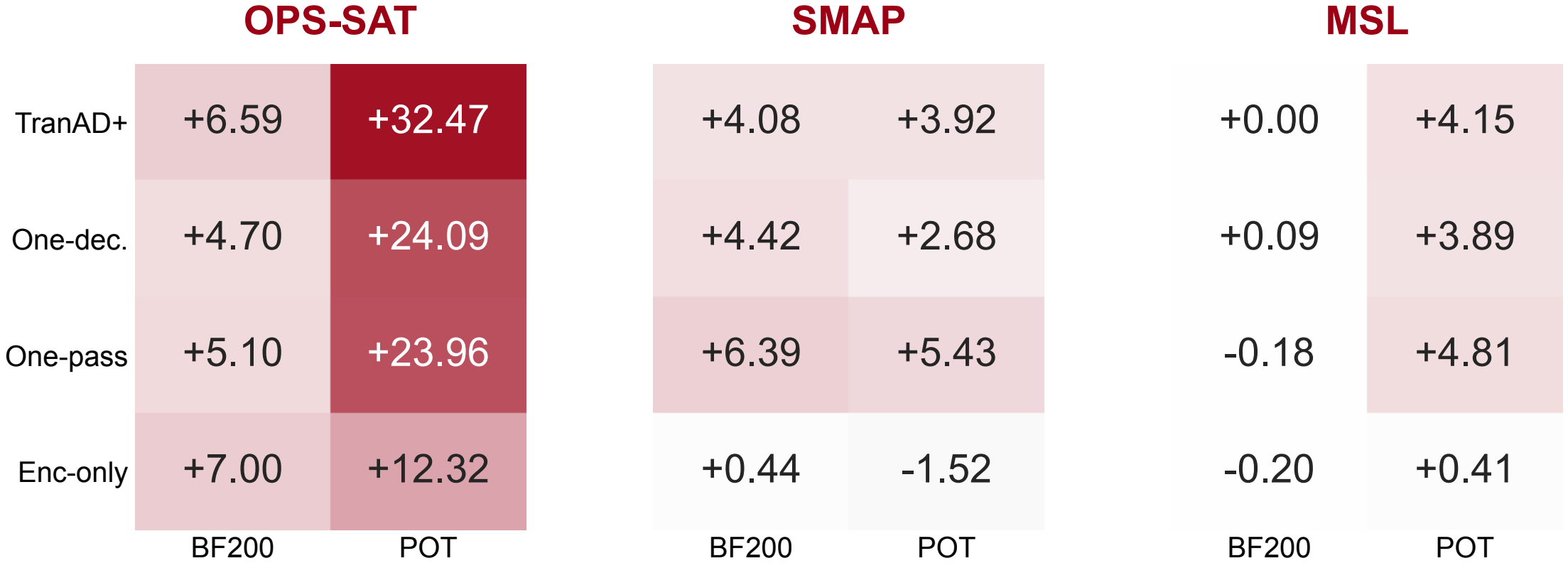
PA · mean over 80 parents per dataset: 4 architectures × 4 W/FF settings × 5 seeds

Dataset	Method	FP32 (%)	INT8 (%)	INT4 (%)
OPS-SAT	BF200	85.55	87.52	91.40
OPS-SAT	POT	59.88	62.23	83.09
SMAP	BF200	81.38	81.47	85.22
SMAP	POT	79.82	79.77	82.45
MSL	BF200	46.71	46.70	46.64
MSL	POT	85.91	85.88	89.22

Integer matrix products; threshold and alpha reselected for each precision.

INT4 · architecture-dependent effects

INT4 minus matched FP32 · PA · mean paired change over 4 configurations × 5 seeds



The gain depends on architecture and threshold selection; encoder-only is not always favoured.

INT4 · PA and pointwise results

F1 (%) · same PA-selected alpha and threshold · mean over 80 parents per dataset

Dataset	Method	PA FP32	PA INT4	No PA FP32	No PA INT4
OPS-SAT	BF200	85.55	91.40	14.53	9.03
OPS-SAT	POT	59.88	83.09	4.35	4.15
SMAP	BF200	81.38	85.22	5.86	4.93
SMAP	POT	79.82	82.45	3.33	3.33
MSL	BF200	46.71	46.64	5.80	5.77
MSL	POT	85.91	89.22	15.41	11.96

The no-PA columns are not independently optimized pointwise F1 results.

Keeping FP32 calibration fixed

Original FP32 alpha and threshold kept fixed · PA · mean paired F1 change (pp)

Dataset	BF INT8	BF INT4	POT INT8	POT INT4
OPS-SAT	-0.90	-9.95	+2.44	+20.10
SMAP	-0.86	-2.94	-0.36	-0.55
MSL	-0.08	-0.12	+0.12	-20.07

INT4 can change score calibration substantially; a previous threshold need not transfer.

Model representation size

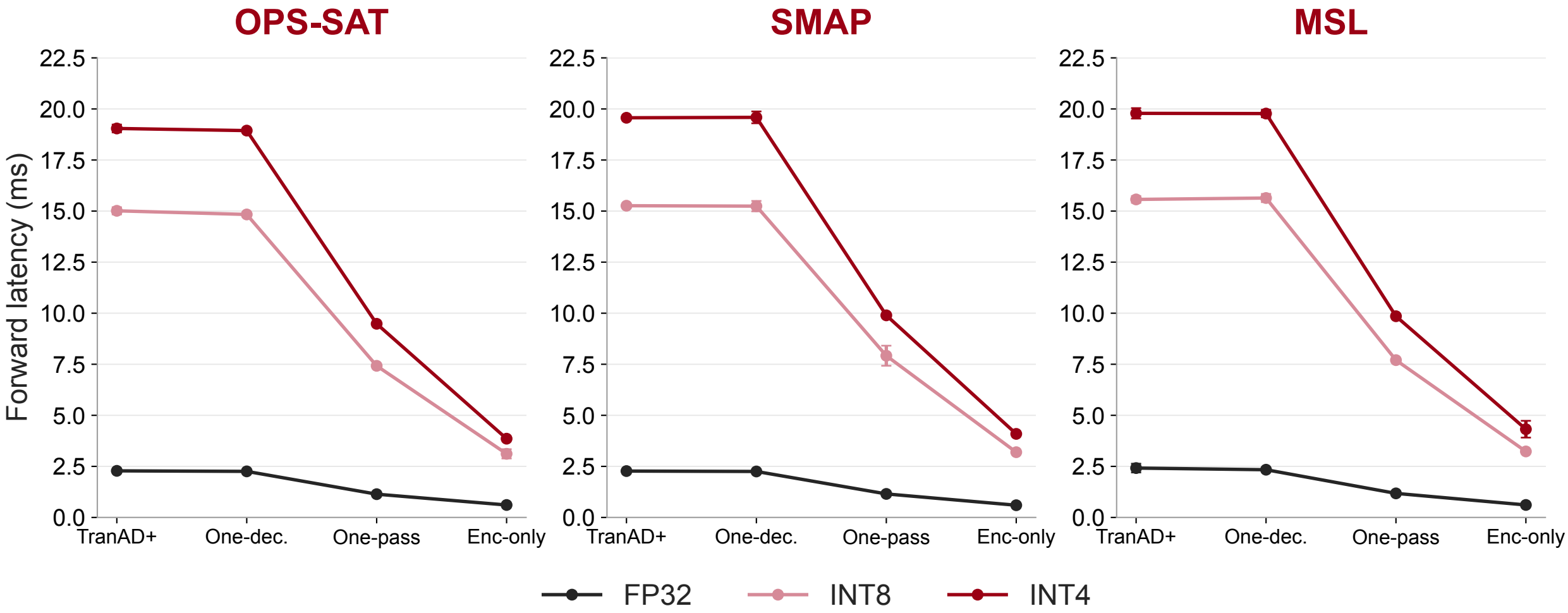
Reference W/FF · model tensor payload including biases, scales and positional buffers

Dataset	Model	FP32 (KiB)	INT8 (KiB)	INT4 (KiB)
OPS-SAT	TranAD+	1.05	0.81	0.72
OPS-SAT	Encoder-only	0.35	0.28	0.25
SMAP	TranAD+	265.92	77.00	44.65
SMAP	Encoder-only	66.70	20.36	12.43
MSL	TranAD+	1285.41	357.39	200.83
MSL	Encoder-only	331.50	103.88	65.47

Model payload reduction is not the same as total peak inference-memory reduction.

Measured inference latency

A100 · batch 1 · 100 warm forward measurements · mean ± SD across 5 model seeds



This integer backend saves model storage but is slower than matched FP32 in every measured case.